MILITARY SPECIFICATION SHEET

PAINT, EPOXY-POLYAMIDE, TOPCOAT, RED, FORMULA 156R66, TYPE II

This specification is approved for use within the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the paint described herein shall consist of this document and the latest issue of MIL-P-24441(SH).

FORMULA. This formula covers red epoxy-polyamide paint designated Navy Formula 156R66, type II for interior or exterior use. The paint shall consist of the ingredients specified in the quantities specified.

Component A

	Pounds
Thixotrope[1/]	-
Polyamide[2/]	20
Polyamide adduct[3/]	320
Magnesium silicate[4/]	250
Titanium dioxide[5/]	100
Yellow iron oxide[6/]	50
Red iron oxide[7/]	300
Paint thinner[8/]	143
2-ethoxy ethanol[9/]	119

Component B

	Pounds
Thixotrope[1/]	-
Epoxy resin[10/]	586
Magnesium silicate[4/]	50
Diatomaceous silica[11/]	150
2-butoxy ethanol[12/]	54
2-ethoxy ethanol[9/]	174

See footnotes on next page.

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QUANTITATIVE REQUIREMENTS. The paint shall meet the following quantitative requirements and the qualitative requirements of section 3 of the general specification. Components A and B shall be mixed 1:1 by volume for mixed components tests. Tests shall be performed in accordance with the general specification.

Requirements	Component A		Component B		Mixed component	
	Min	Max	Min	Max	Min	Max
Pigment content, percent	51.8	55.8	18.5	22.5	---	---
Volatiles, percent	24.3	28.3	20.3	24.3	---	---
Nonvolatile vehicle, percent	17.9	21.9	55.2	59.2	---	---
Water, percent	---	0.5	---	0.5	---	---
Coarse particles, percent	---	0.3	---	0.3	---	0.5
Consistency, grams	140	260	120	205	130	235
Pounds per gallon	12.6	13.0	9.8	10.2	11.2	11.6
Set-to-touch time, hours at 40°F	---	---	---	---	---	2
Set-to-touch time, hours at 73°F	---	---	---	---	---	2
Dry-hard, hours at 40°F	---	---	---	---	---	24
Dry-hard, hours at 73°F	---	---	---	---	---	6
Fineness of grind, NS	3	---	3	---	3	---
Flash point, °F	90	---	110	---	95	---
Titanium dioxide, percent of pigment	12.5	---	---	---	---	---
Pot life, hours at 73°F	---	---	---	---	5	---
Gloss, 60 degrees specular, percent	---	---	---	---	25	---
Sag resistance, mils	---	---	---	---	6	---
Color, dry film to approximate the standard color chip 13/	---	---	---	---	Conform	
Contrast ratio, 3 mils dry film	---	---	---	---	0.98	---

1/ The amount and type of thixotrope shall be selected by the manufacturer to meet all requirements of the general specification and this specification sheet.
2/ GENAMID 2000, Henkel Corporation; UNIREZ 2810, Union Camp; AZAMIDE 600, AZ Products; ANCAMINE 507, Pacific Anchor Chemical Corp.; EPOTUF SF7791, Reichhold Chemical; TriChem 9200, Trimont Chemicals.
3/ VERSAMID 280B75, Henkel Corporation; UNIREZ 2180B75, Union Camp; AZAMIDE 680B75, AZ Products; ANCAMINE 700B75, Pacific Anchor Chemical Corp.; EPOTUF SF7792, Reichhold Chemical; TriChem 9280-B-75, Trimont Chemicals.
4/ Pfizer Talc MP 30-36; Cyprus Montana Talc, Mistron 400.
5/ DuPont Tipure R960; NL Chemicals, TX 2101; Kerr-McGee, Tronox Tr 822.
6/ Pfizer Y03587.
7/ Pfizer RO-6097.
8/ Chevron 350H, VARSOL-18, AMSCO Mineral Spirits.

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9/ 99 percent pure by gas chromatography, density 0.93, refractive index (68°F) = 1.407, IR spectrum characteristic of compound.
10/ EPON 828, Shell Chemical Co.; ARALDITE 6010, Ciba-Geigy; DER 331, Dow Chemical Co.; AZEPOXY 128, AZ Products; EPO-TUF 37-140, Reichhold Chemical; EPI-REZ 510, Celanese; TriChem 728, Trimont Chemicals.
11/ Celite 281 or 499, Johns-Manville; MICRO-KEN 801 and MICRO-KEN 803.
12/ 99 percent pure by gas chromatography, density 0.90, refractive index (68°F) = 1.418, IR spectrum characteristic of compound.
13/ Use FED-STD-595 Color Chip-Color No. 20109. Color shall approximately match the color chip.

Use of alternate ingredients in this formula must have prior approval by the Naval Sea Systems Command. Approval will be based on review of data showing equivalent physical and chemical characteristics to the specified ingredient. It will be necessary to demonstrate that paint made using the ingredient will conform to all requirements of this specification.

Revision letters are not used to denote changes due to the extensiveness of the changes.

Preparing activity:
Navy - SH
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